ETH zürich

Mobile Computing

Emerging Technologies
6G LiFi Visible Light Communication

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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

VLC History - Photophone

- First dated application of VLC
- Invented by Alexander Graham Bell, 1880
- World’s first wireless telephone
- ~ 200 meters, by modulating sunlight

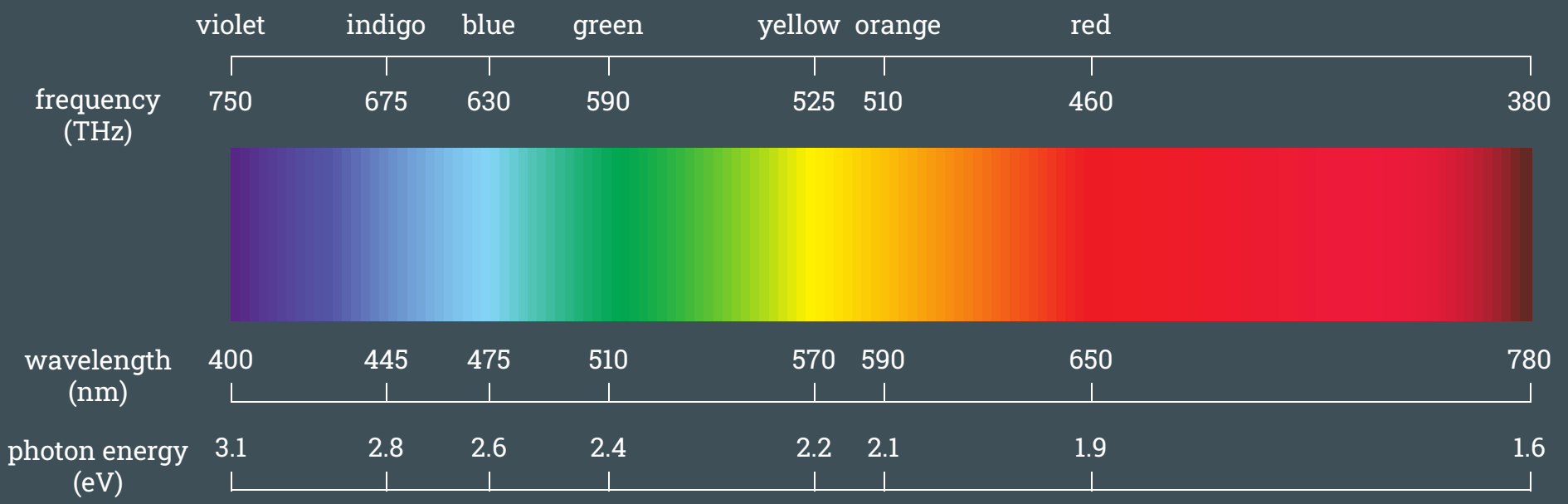
VLC History - Photophone



Why Visible Light?

- The visible light spectrum is a broad (400 THz) and unlicensed spectrum
  - Visible light is not perceived as radio wave, or harmful
- LEDs are already embedded in many consumer electronics devices, and (of course) in lights
- Underwater communication works best with visible light
- VLC is considered for next generation cellular (“6G”) standards, and for future Wi-Fi generations (“Li-Fi”)

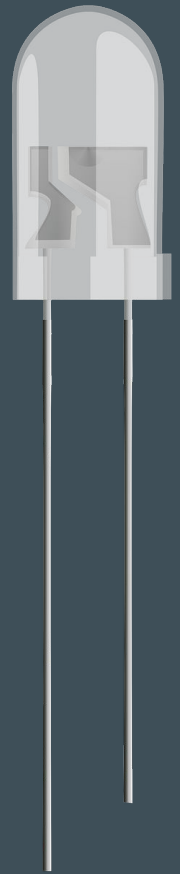
Visible Light Spectrum (> 400’000 GHz)



THz = 10^{12} cycles per second · nm = 10^{-9} metre · eV = electron volt

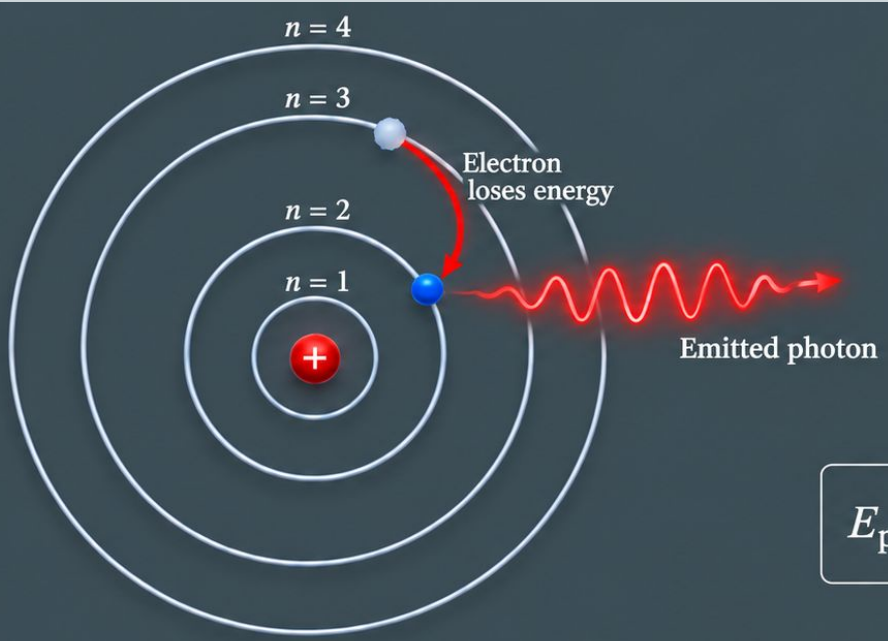
Adapted from Encyclopaedia Britannica (2012); values approximate.

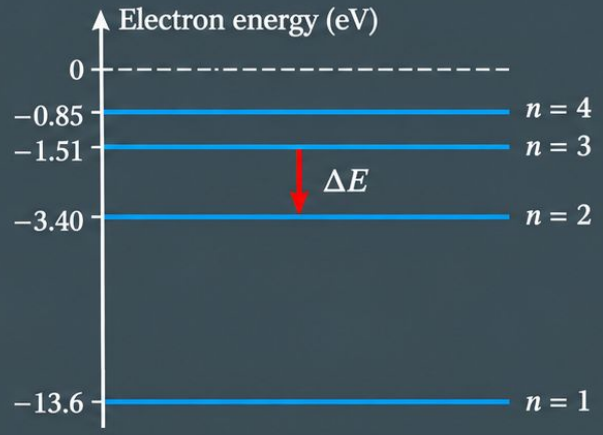
https://www.britannica.com/science/light





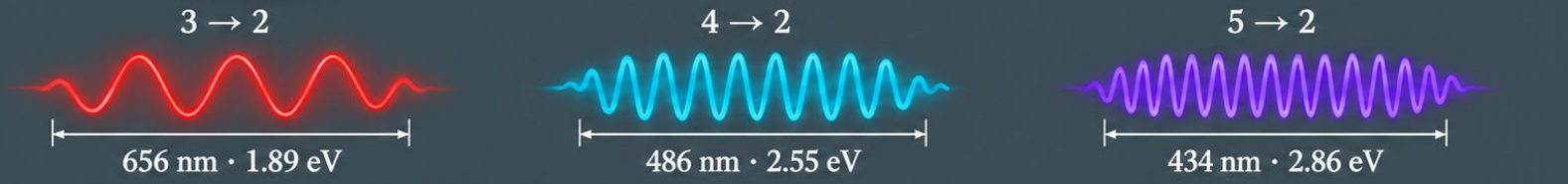
Photon Emission: Energy, Wavelength, and Color





$$E_{\text{photon}} = \Delta E = E_{\text{initial}} - E_{\text{final}} = hf = \frac{hc}{\lambda}$$

Larger energy gap → shorter wavelength





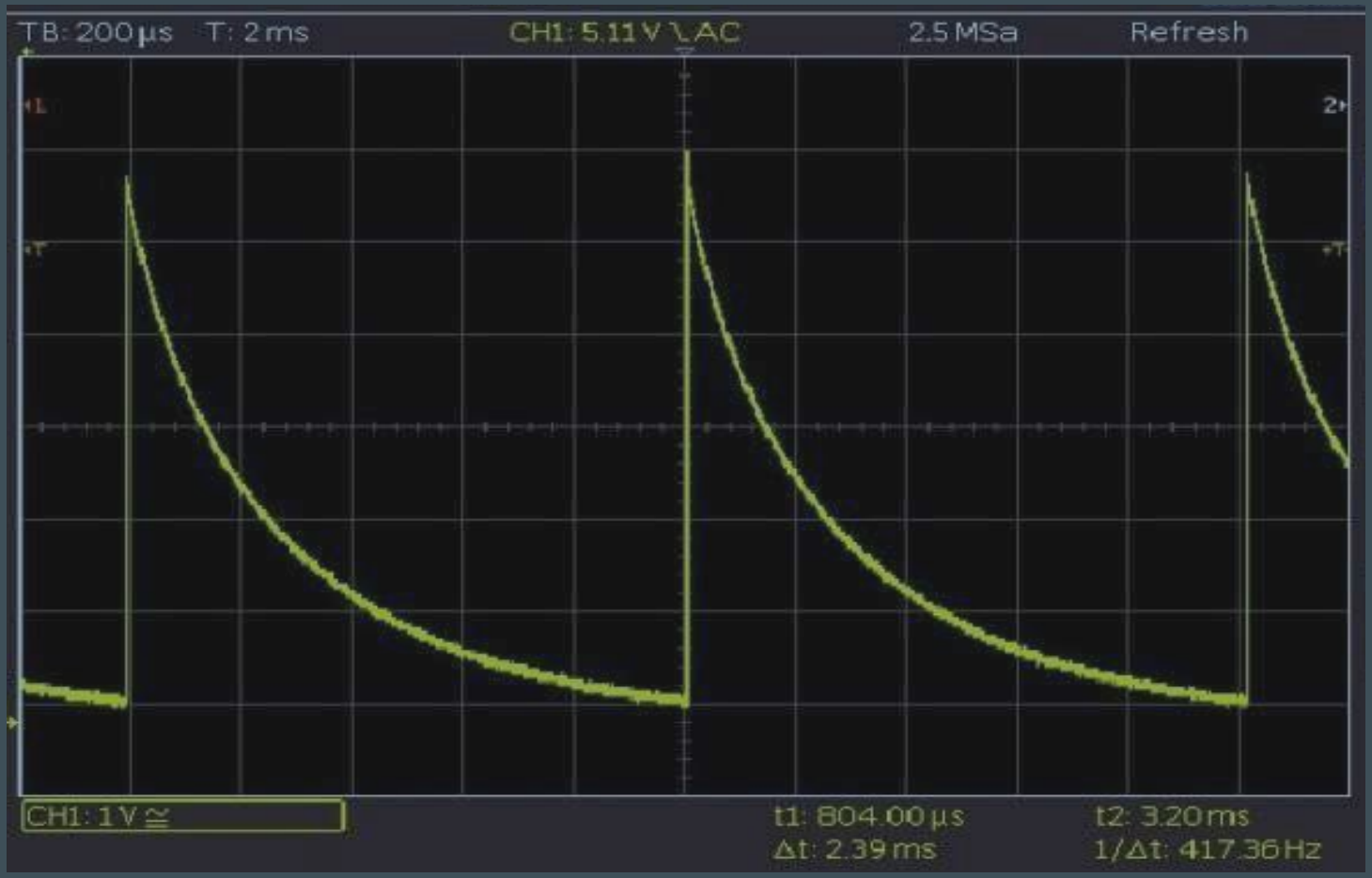
Shorter wavelength / higher photon energy

Longer wavelength / lower photon energy

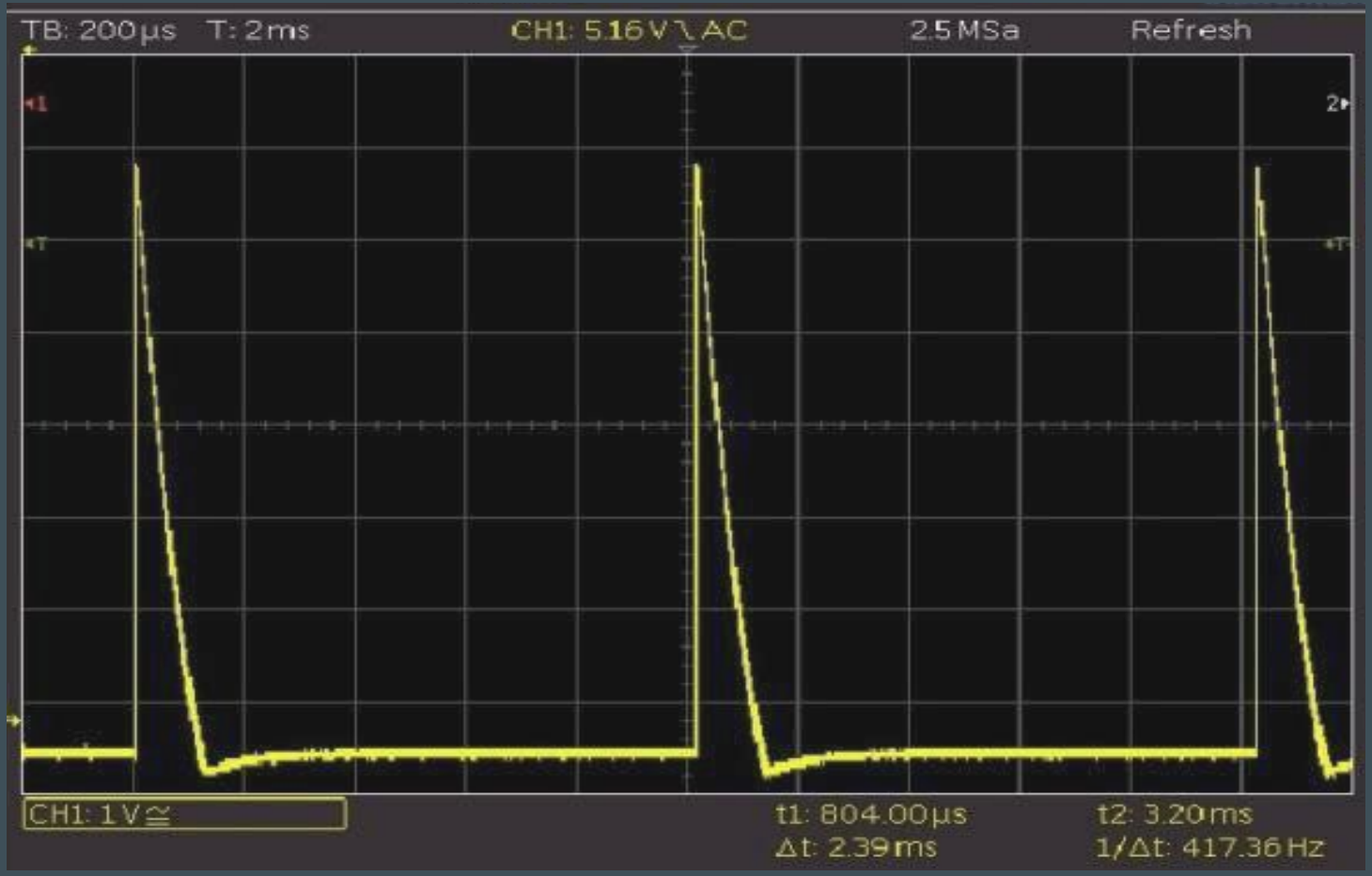
LEDs as Low Cost Receivers

- LEDs can be used as light receivers, like photodiodes, but with less sensitivity
  - P. Dietz, W. Yerazunis, and D. Leigh: “Very low-cost sensing and communication using bidirectional LEDs,” UbiComp 2003.
  - Photons create an electric current that is proportional to the intensity of the incoming light
  - Reduces device complexity
- One single LED is enough to transmit and receive data
  - Protocol to control the LED is required
  - Tradeoff: low sensitivity, shorter distances, low data rates

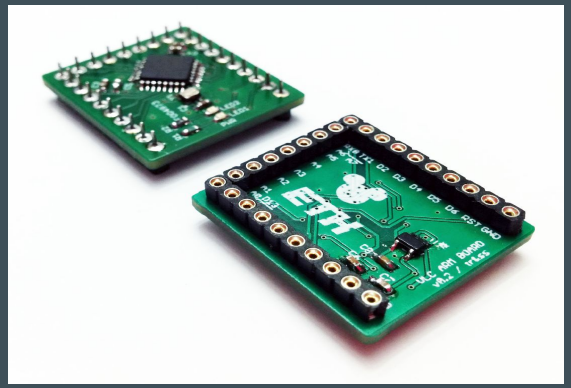
Light Sensing with LEDs (no incoming light)



Light Sensing with LEDs (incoming light)



Low-Cost Software-Defined VLC

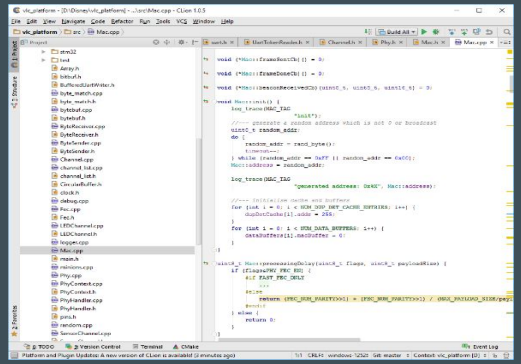


Microcontroller

+

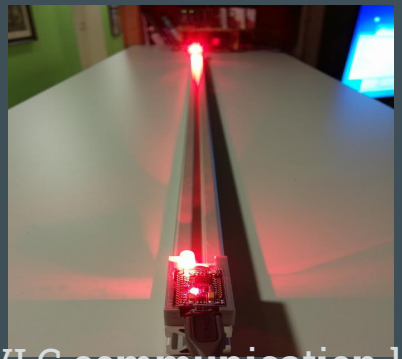


Sender/Receiver



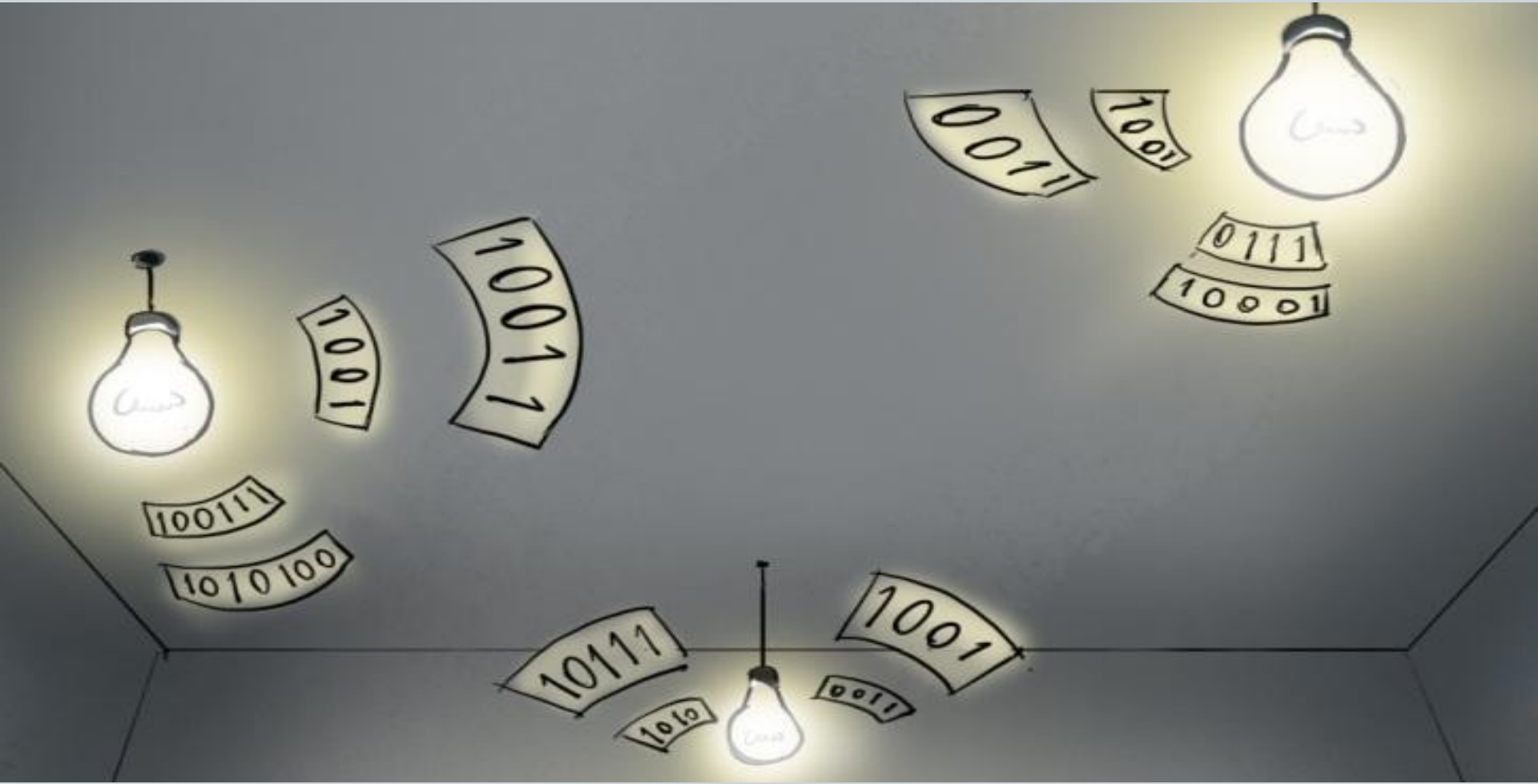
Embedded Software

=

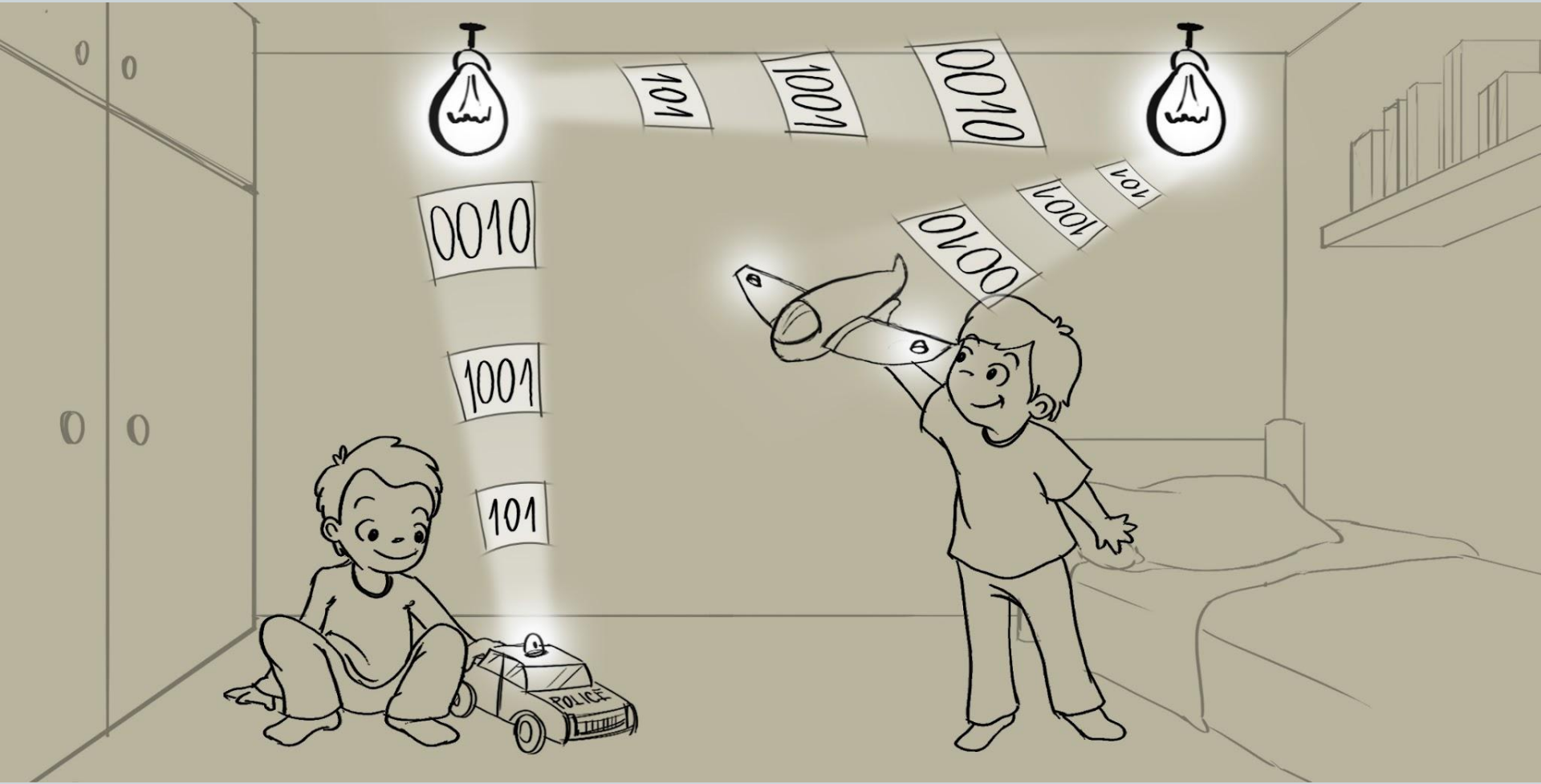


VLC communication link:
1 kb/s over 2 meters

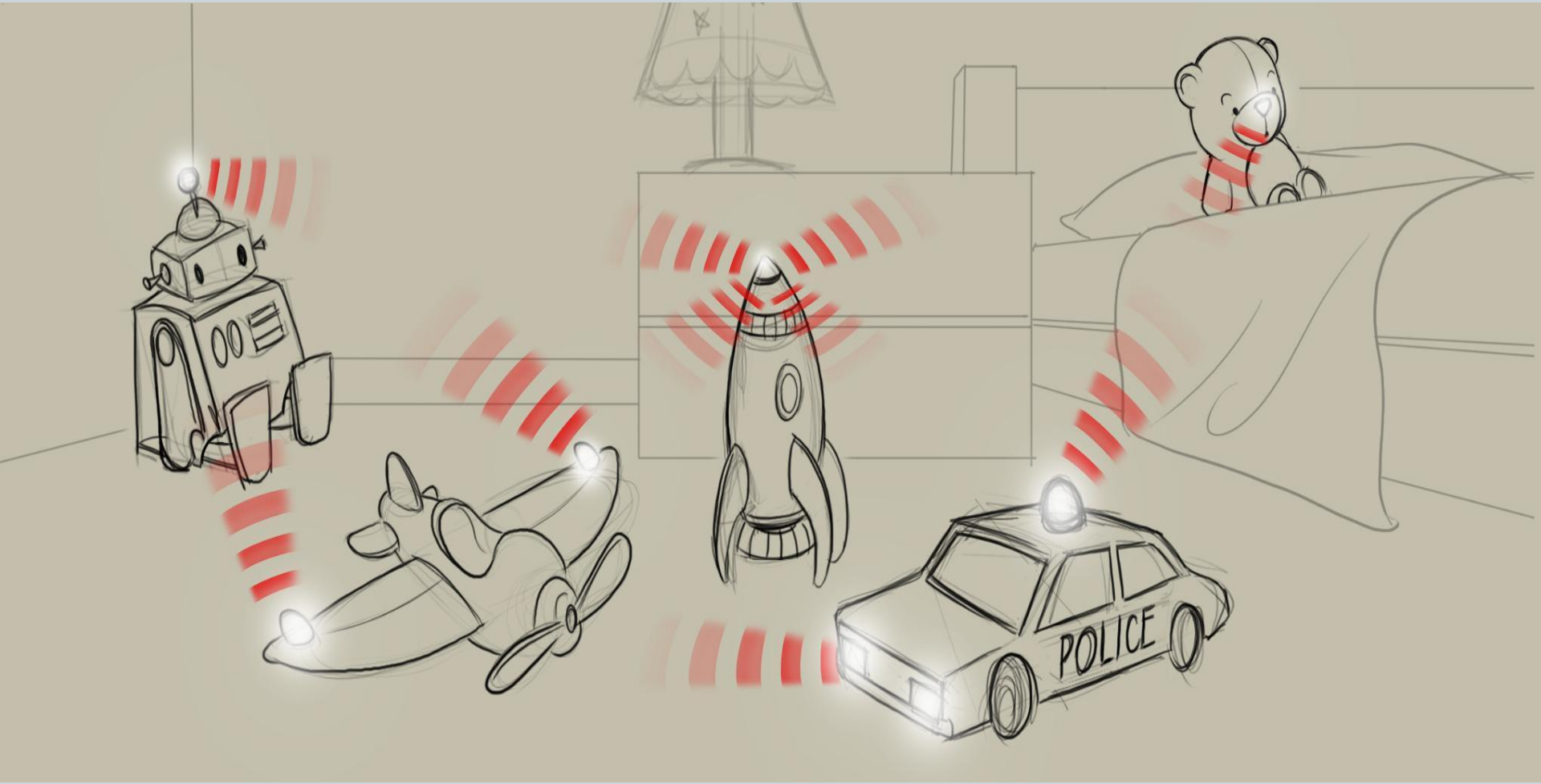
Applications



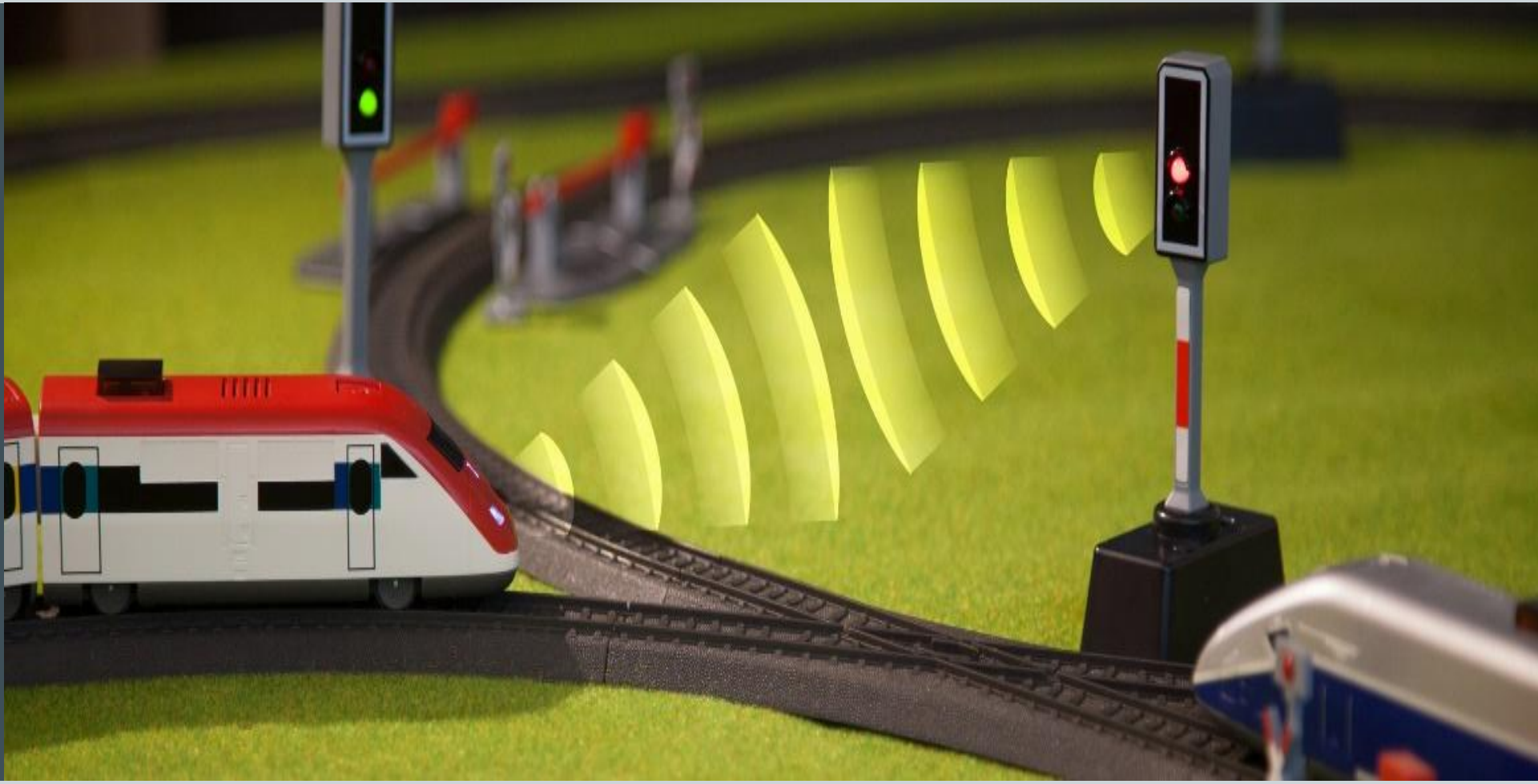
Home Networks, IoT



Toy Networks (remote controlled toys)



Low Cost Model Train



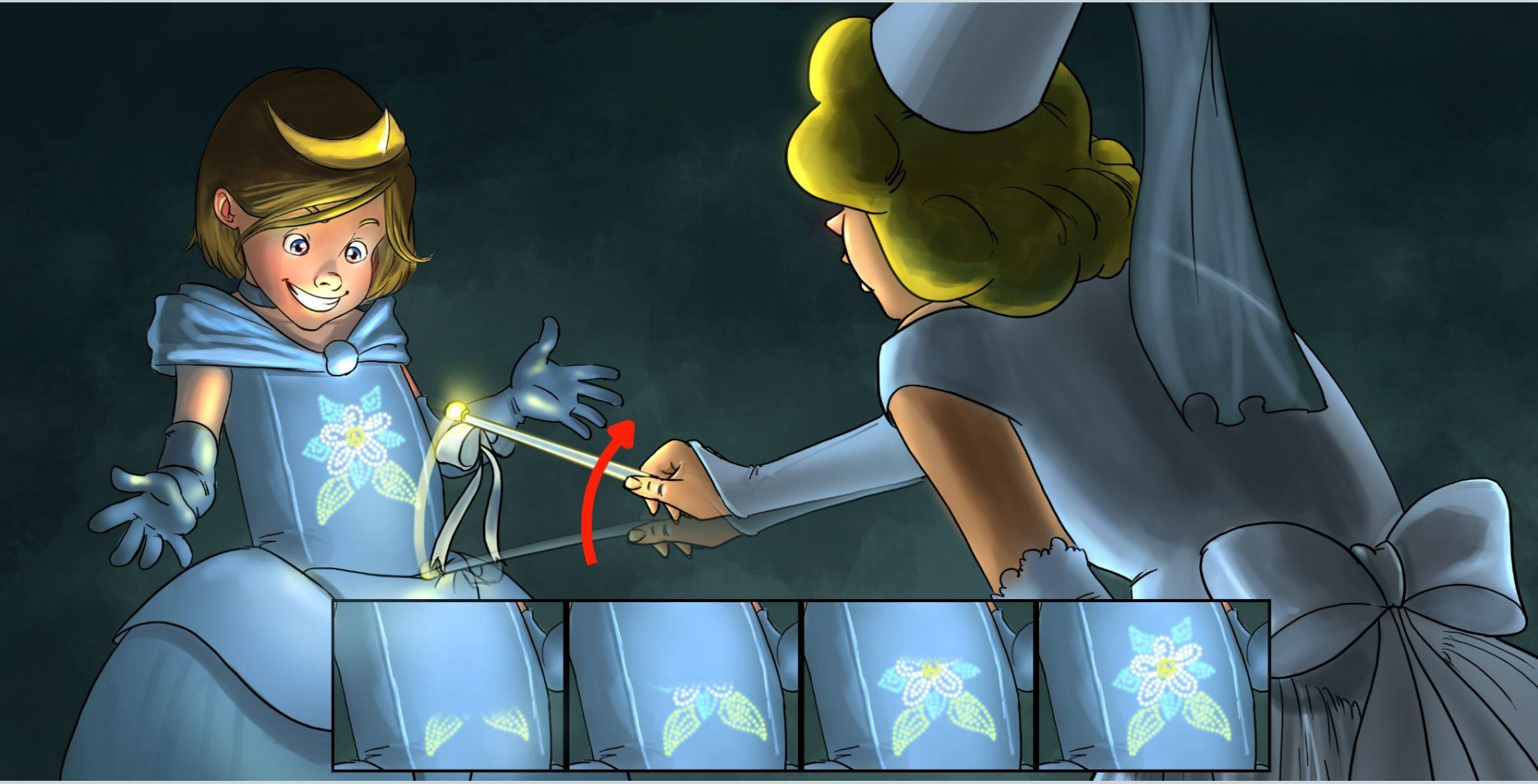
Interactive Toys



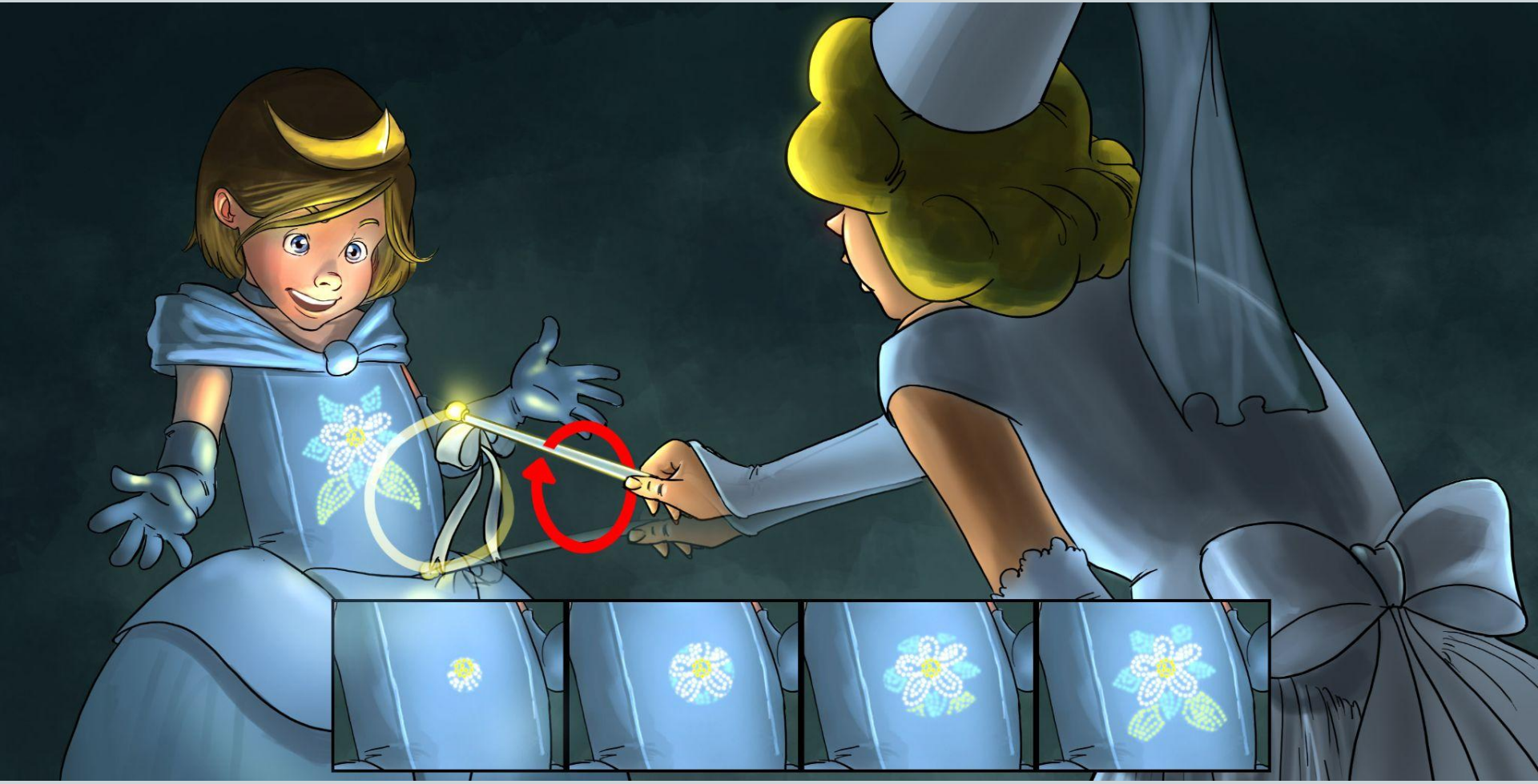
Fashion and Fabrics - Princess Dress



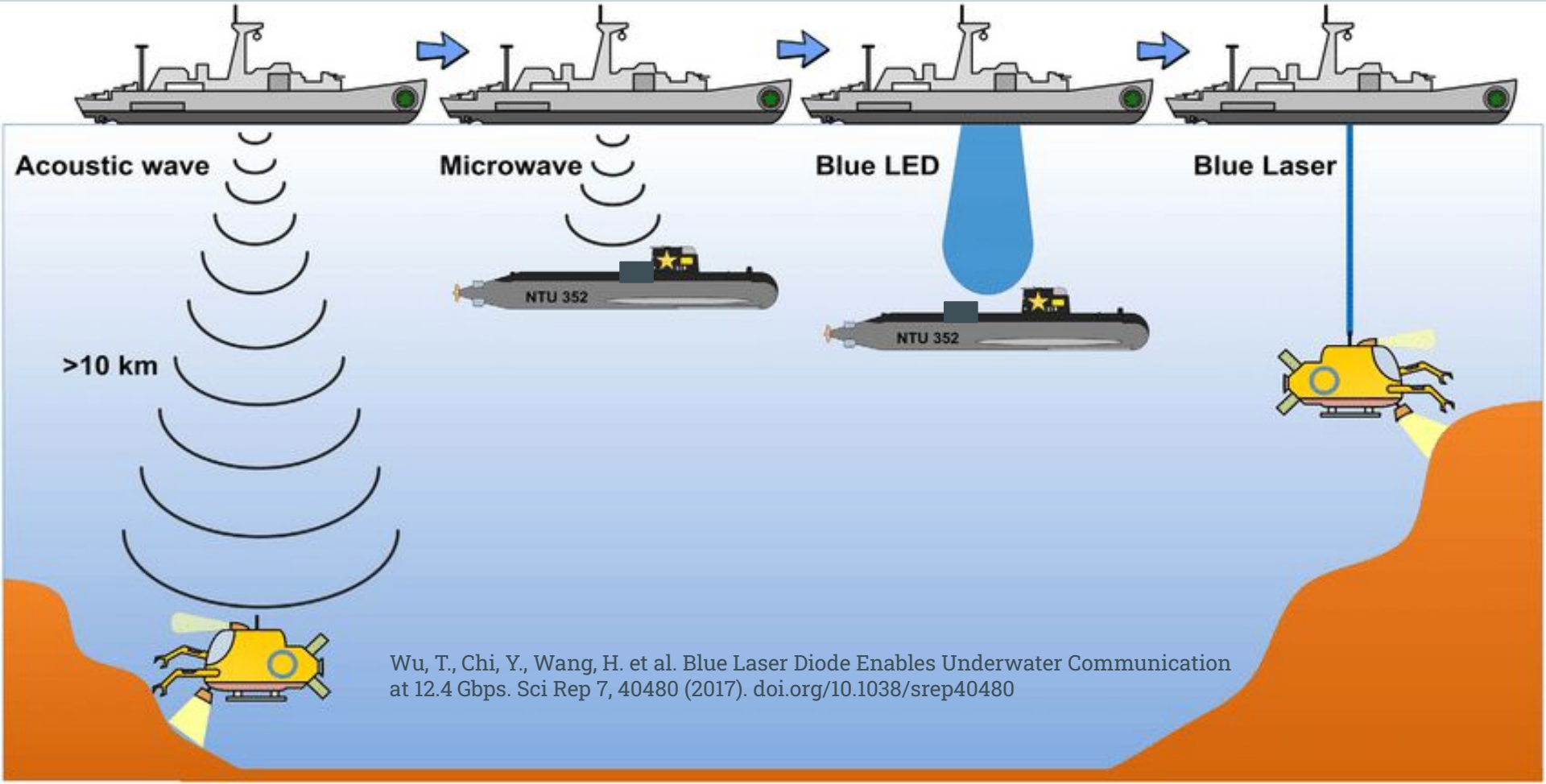
Fashion and Fabrics - Princess Dress



Fashion and Fabrics - Princess Dress



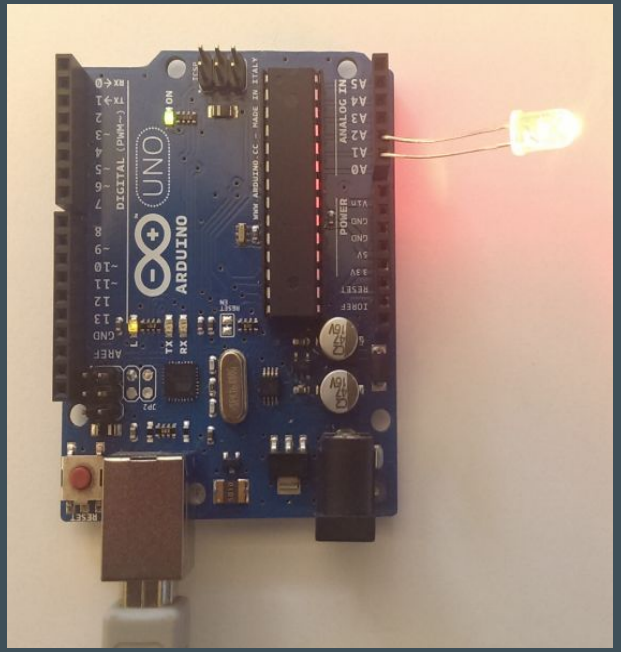
Underwater Communication



Wu, T., Chi, Y., Wang, H. et al. Blue Laser Diode Enables Underwater Communication at 12.4 Gbps. Sci Rep 7, 40480 (2017). doi.org/10.1038/srep40480

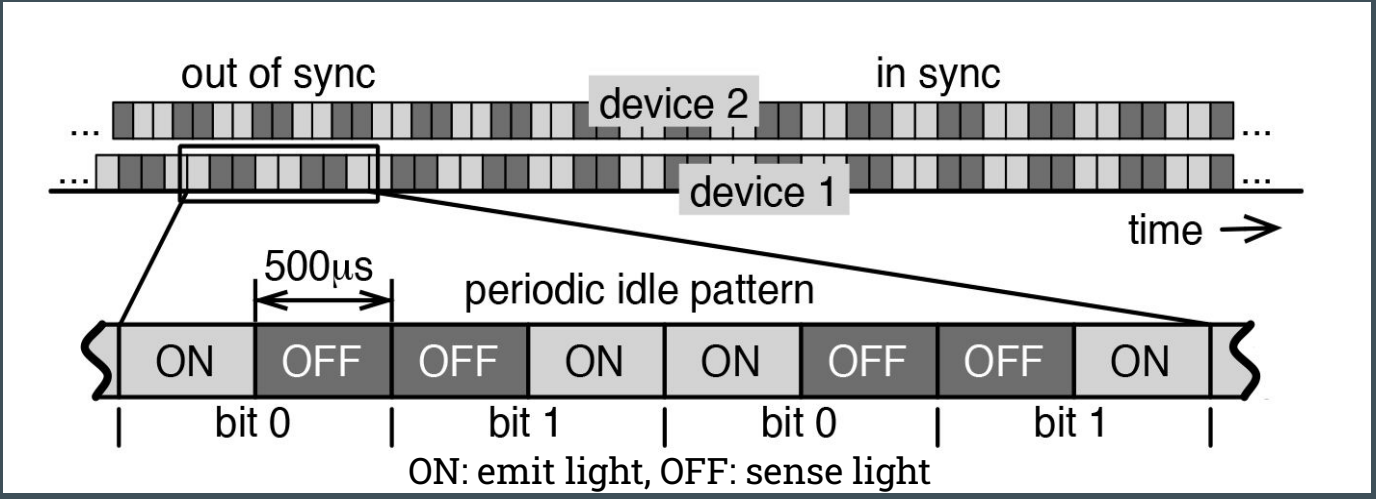
Software-Defined PHY-Layer

- Responsible for:
  - Frame delivery over free-space optical medium
  - Synchronization
  - Medium idle/busy sensing
- Hardware parts:
  - Low-cost 8-bit microcontroller (µC)
  - Standard LED
  - PHY-layer is part of the µC's firmware
- LED is used to sense/emit light (directly connected to µC)



Idle Mode

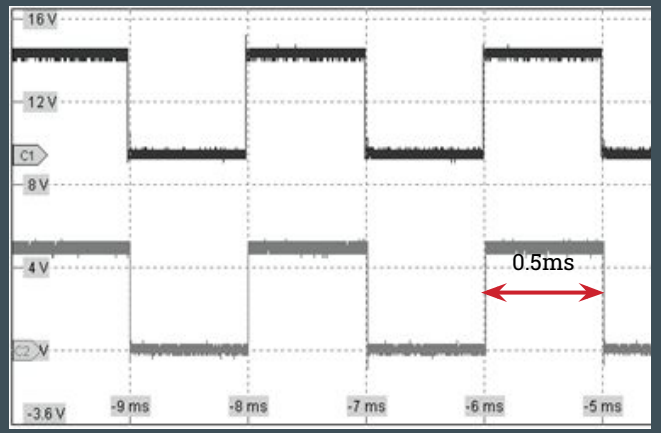
- LED appears to be always on
  - Required for all communication modes (idle, receiving, etc.)
  - Necessary to provide constant brightness (illumination, toys)
- Periodic idle pattern



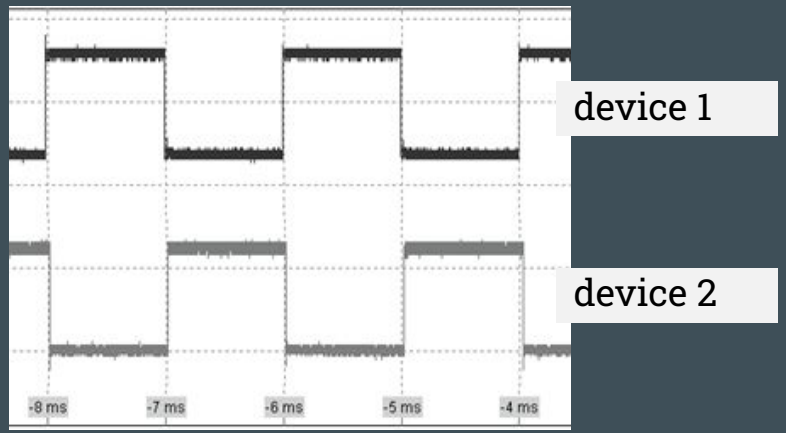
Synchronization During Idle Mode

- Communication requires synchronization
  - Devices have to synchronize to idle pattern, continuously
  - Synchronize so that OFF-slots are aligned
- Enables CSMA/CA MAC protocol

in-sync: aligned OFF-slots



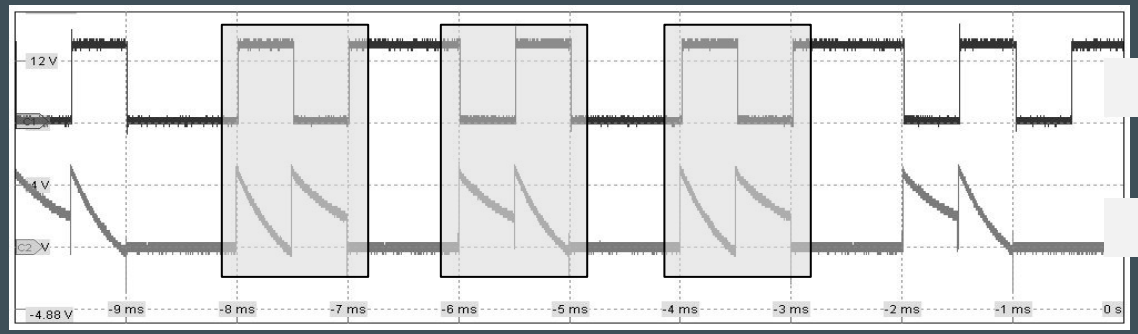
out of sync: OFF aligned with ON slots



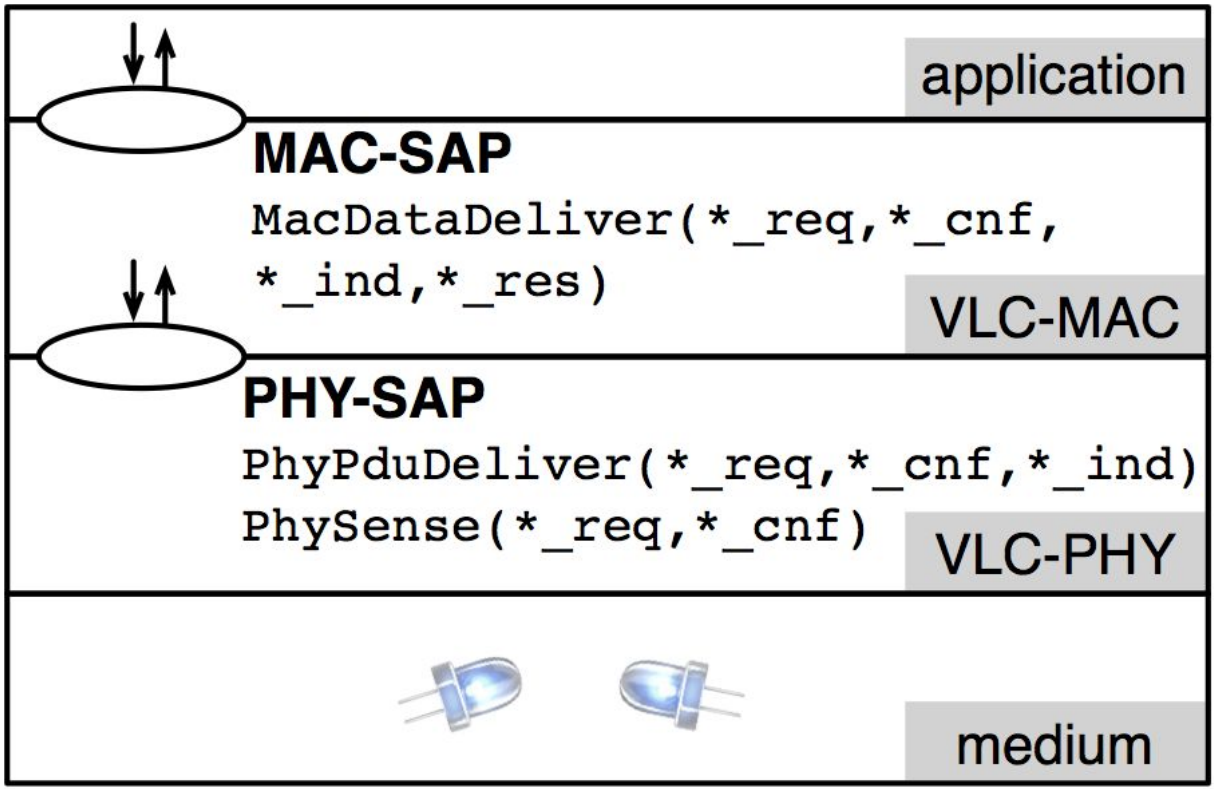
Transmitting (TX) / Receiving (RX) Mode

- TX mode:
  - 2-pulse position modulation (2-PPM)
  - Same number of ON and OFF slots during transmission
- RX mode:
  - Reverse-biased LED, measuring cathode voltage with a microcontroller's Analog-to-Digital Converter (ADC), at the end of the slot

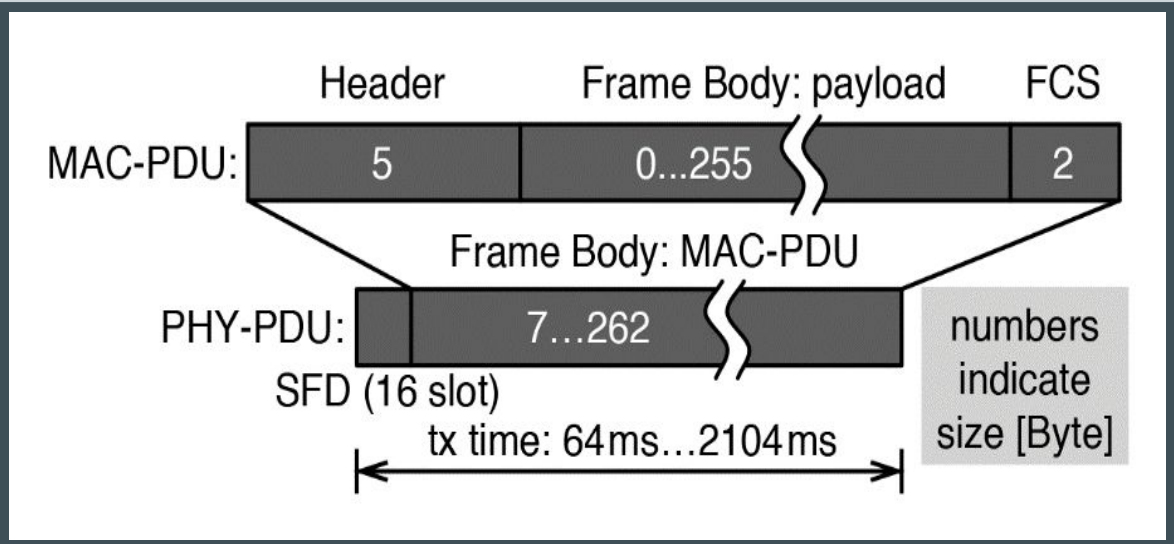
bit	first symbol	second symbol
0	ON	OFF
1	OFF	ON



VLC Reference Model



MAC Protocol Data Unit



MAC- and PHY-PDU definition

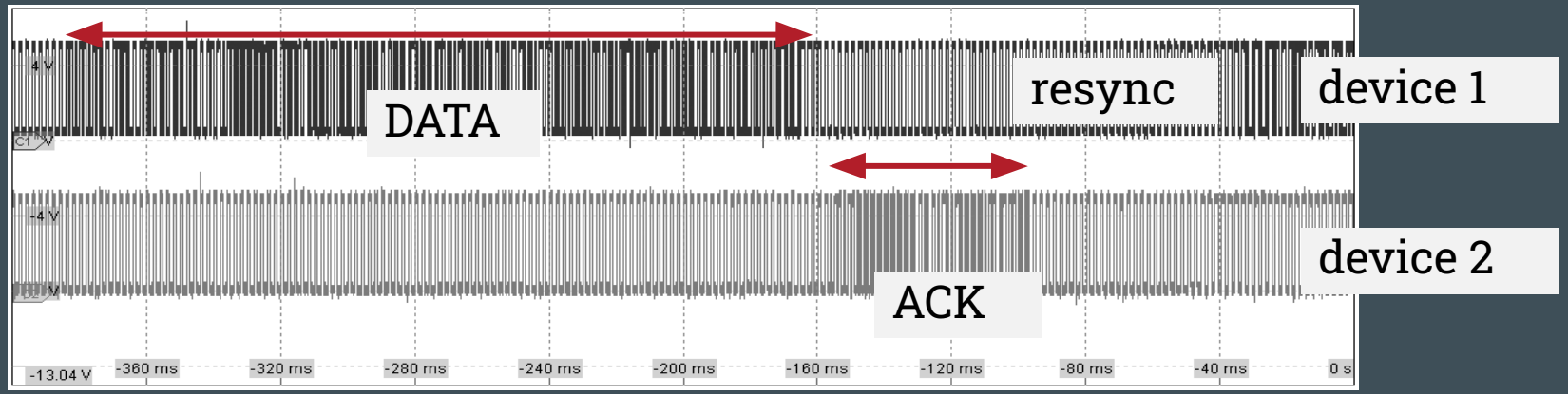
Element	Bytes (Slots)
Header	5 (80)
Body (DATA)	0-255 (0-4080)
Body (ACK)	0 (0)
FCS	2 (32)

Element	Bytes (Slots)
Body size	1 (16)
Control	1 (16)
Destination	1 (16)
Source	1 (16)
Sequence	1 (16)

MAC frame structure

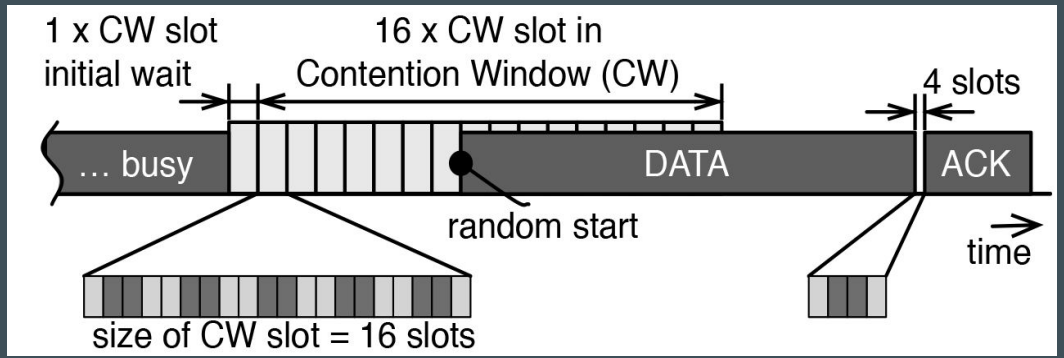
DATA/ACK Frame Exchange

- Device 1 sends a DATA frame to device 2
- Device 2 replies with an ACK (if FCS correct)
- Device 1 continues with the next DATA frame
- Resync in IDLE mode between DATA frames



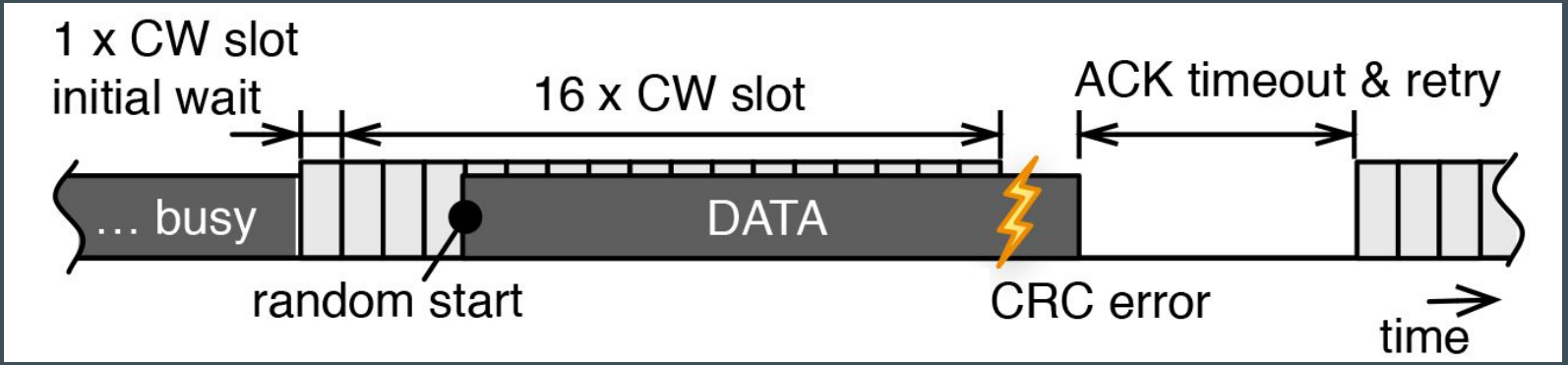
CSMA / CA Medium Access

- ACK has priority, min. contention window: 1 slot
  - Packet transmission starts after random number of CW slots
- When SFD is successfully decoded:
  - Switch to RX mode, prohibit switching to TX mode
- When no frame is decoded, or an SFD is missed:
  - No or small variation in measurements → idle (else: busy)

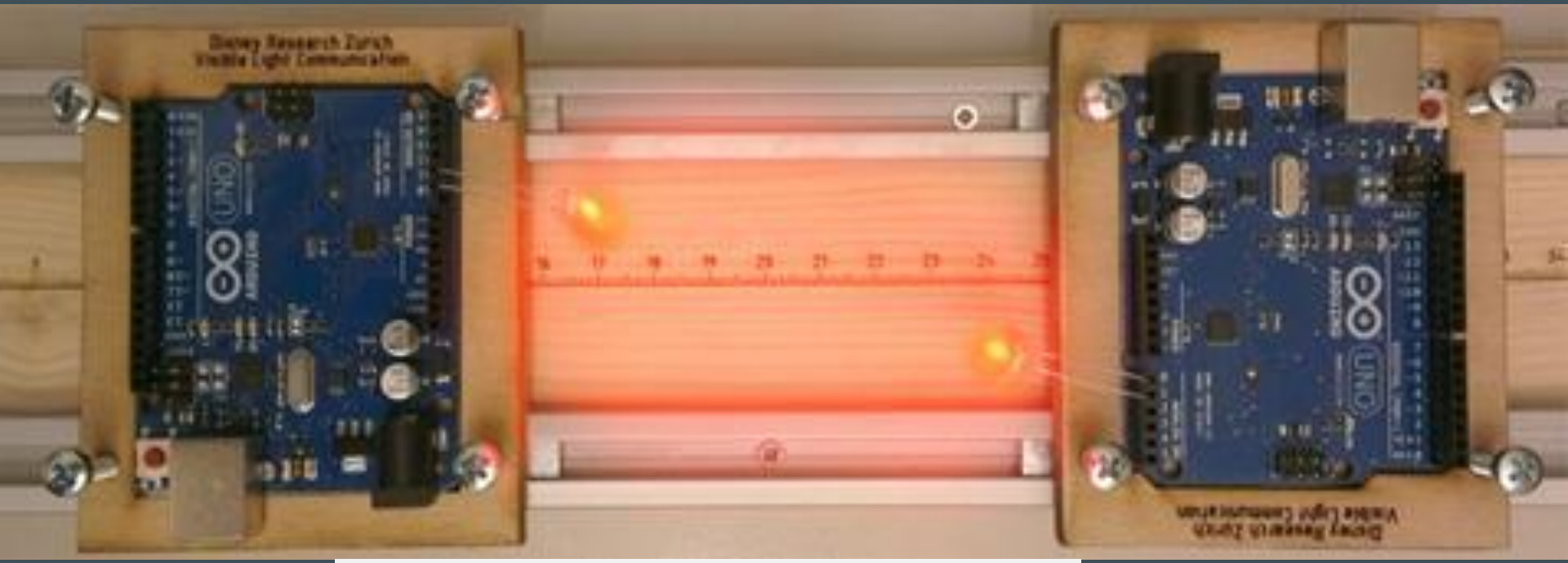


Retransmissions

- On collision or bad reception: CRC error
- Receiver: No ACK is generated
- Transmitter: Retransmit frame after ACK timeout

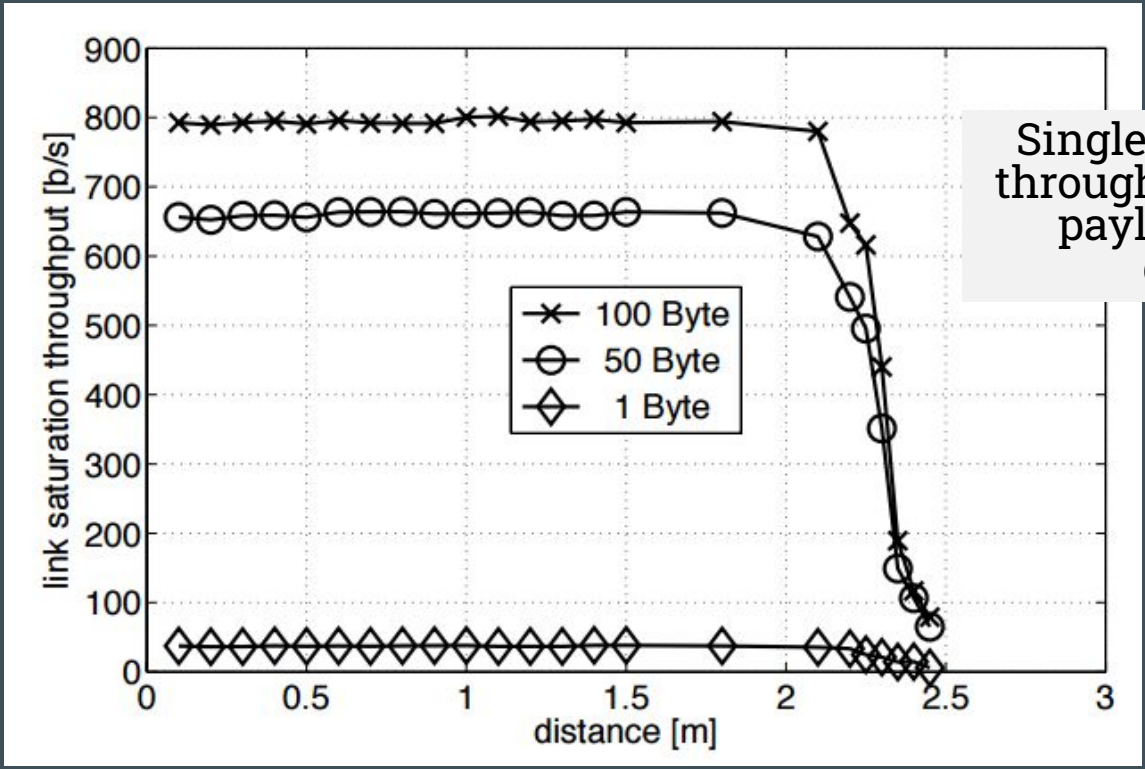


Testbed 1 - Single Link



Testbed for distance measurements,
with boards and LEDs

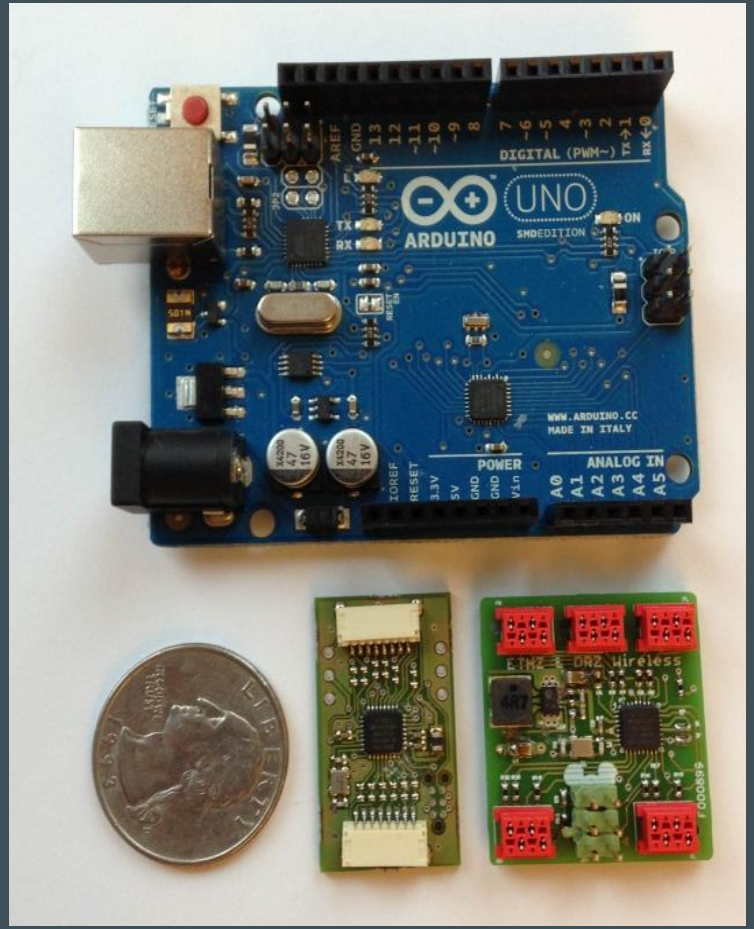
Single Link Throughput (I)



Single link saturation throughput for different payload sizes and distances

Microcontroller Testbed

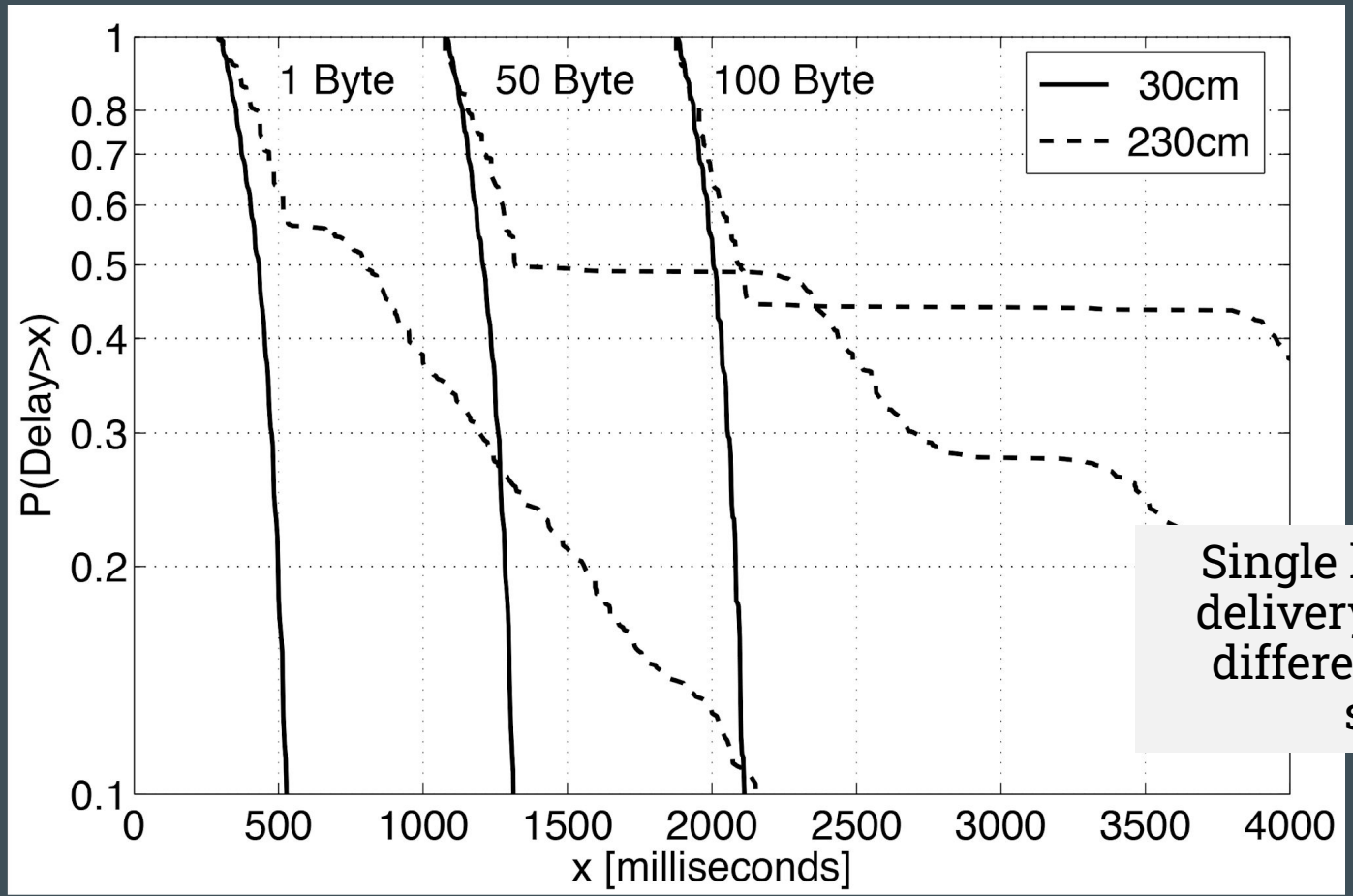
- Arduino boards for prototyping
  - ATmega 328P microcontroller
  - 8 bit, 32K program storage
  - 2K RAM
  - 8 ADC pins (multiplexed)
- ARM



Single Link Throughput (II)

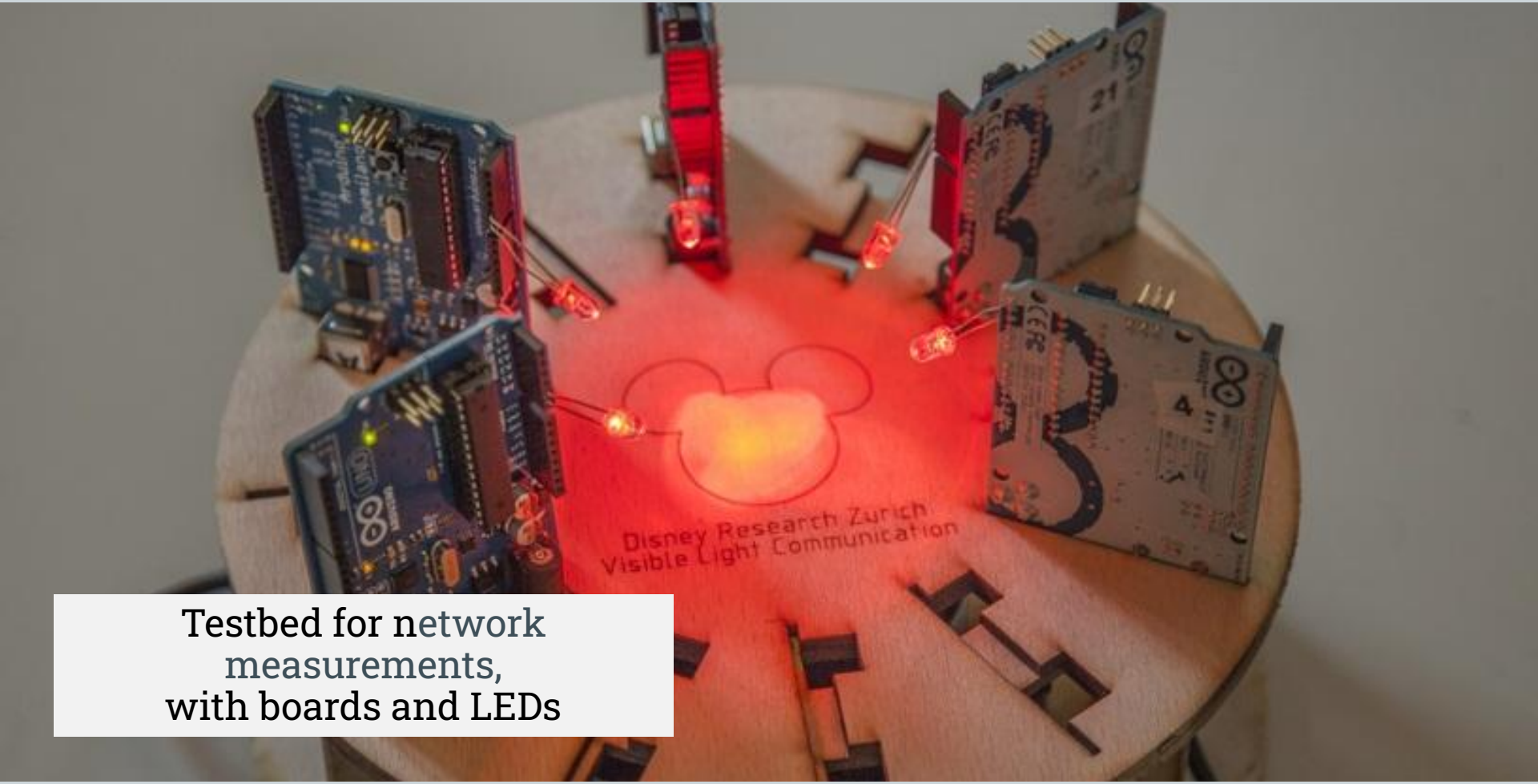
- Distance <= 2 m: Constant throughput for all packet sizes
- Distance > 2 m: Throughput collapses because of increased error probability due to bad reception
- Achieved maximum throughput: 800 b/s (bit per second) for 100 B (100 Byte) packets

Single Link Delays (I)



Single link packet delivery delays for different payload sizes

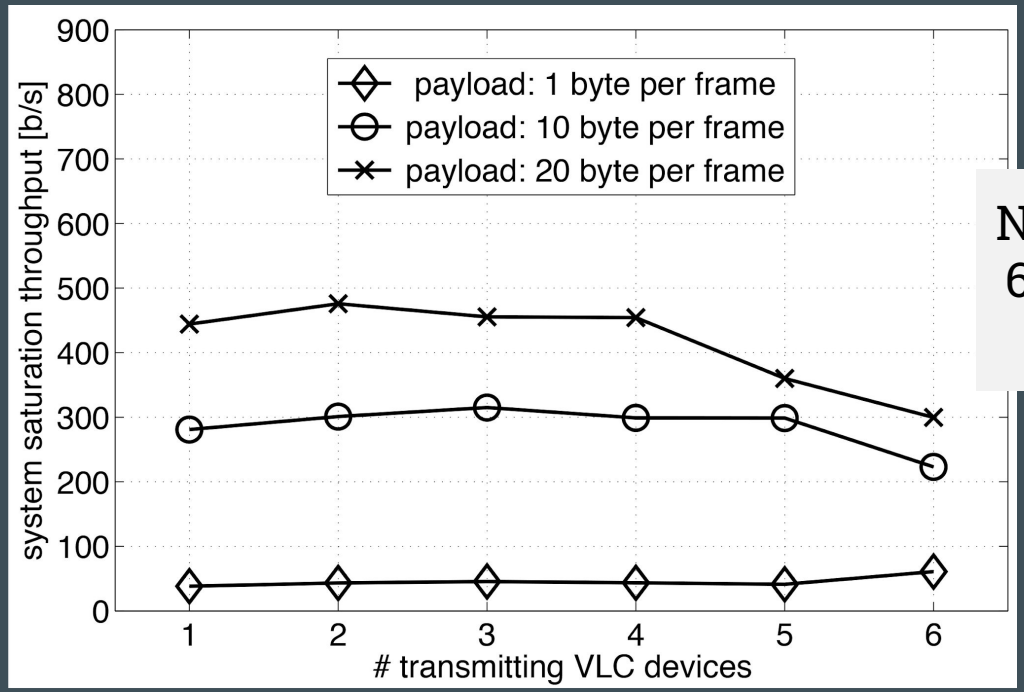
Testbed 2 - Network



Testbed for network measurements, with boards and LEDs

Network Throughput (I)

- Devices transmit data and compete for the medium
- No hidden stations

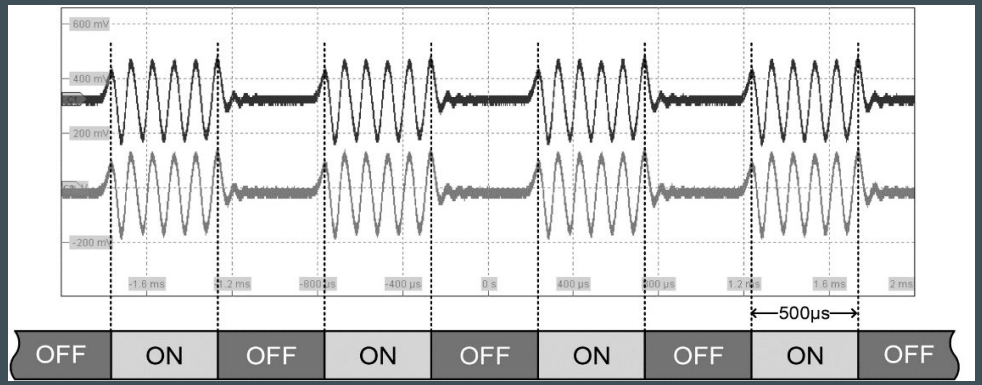


Network throughput for 1 to 6 transmitting devices and different payload sizes

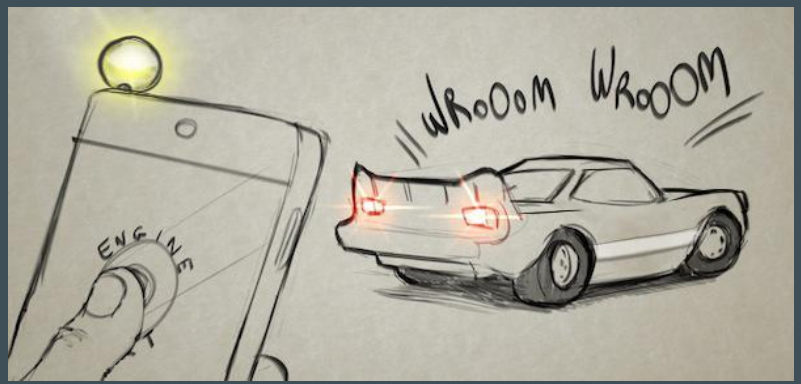
Network Throughput (II)

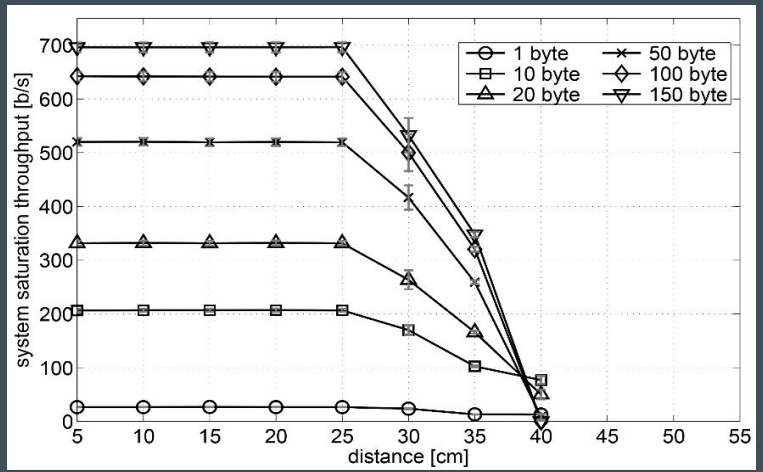
- Smaller packet sizes due to increased error probability
- Less than five transmitting devices: Constant throughput, medium access handled sufficiently by MAC protocol
- More than five transmitting devices: Reduced throughput because of retransmissions due to less stable synchronization
- Network throughput: 500 b/s (bit per second)

From Sound to Sight









Li-Fi

- IEEE 802.11bb standard finished in 2023
  - Up to 9.6 Gb/s
  - Optical Multi-Carrier OFDM
  - Infrared, visible light, ultraviolet
- Products not yet available
  - Market has not developed so far
  - Role of Wi-Fi Alliance not clear



Infrared vs Visible Light

- Differences
  - We see visible light; IR is invisible, felt mostly as heat
  - IR excites molecular vibrations (good for heating/sensing)
  - IR scatters less in dust, smoke, but is absorbed by water/CO_2
- Wavelength / energy
  - Visible ≈ 380…750 nm (higher-energy photons)
  - Infrared (IR) ≈ 750 nm…1 mm (lower-energy)
- Uses
  - Remote controls (940 nm, low power, 38 kHz pulses)
  - Thermal cameras
  - Show control (Events, Theme Parks)
  - Drone swarms

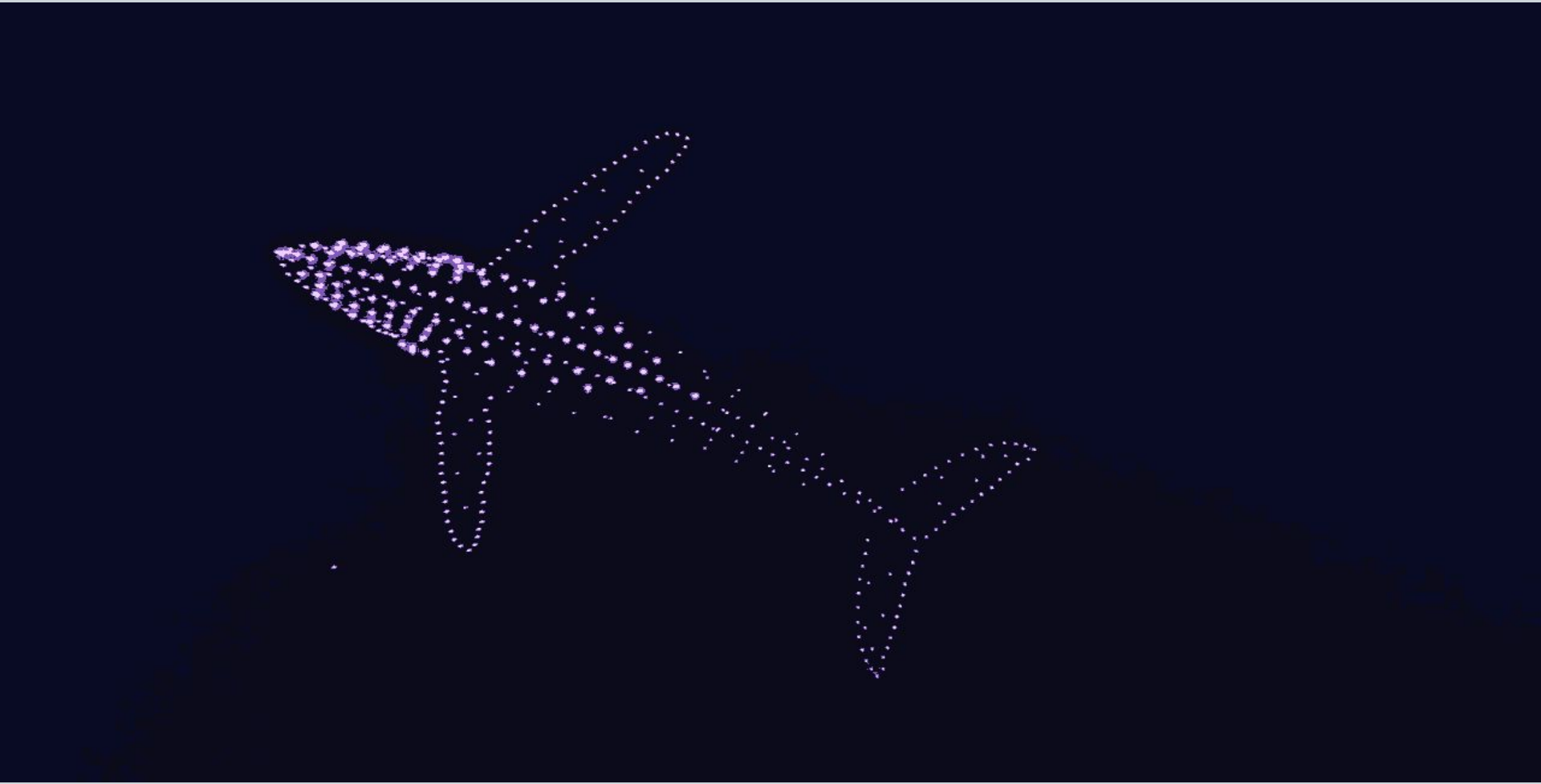
Coldplay LED Bracelet



Disney Theme Park Experience



Drones



Disney Project



Exercise: Visible Light Communication

- Open in your course folder: 02_Notebooks/03_Mobile_Systems/ 03.4_LiFi.ipynb
- Answer the questions and upload the completed notebook to Moodle as part of Homework Assignment 02.

